

Discharge section

Device	Pin	ESP32/Connection	Notes
INA219 (0x40)	VCC	3V3	From ESP32
	GND	GND	Shared
	SDA	GPIO21	I2C data
	SCL	GPIO22	I2C clock
	A0	GND	For 0x40
	A1	GND	For 0x40
	VIN+	Battery +	From battery
	VIN-	Load +	To load
OLED	VCC	3V3	From ESP32
	GND	GND	Shared
	SDA	GPIO21	I2C data
	SCL	GPIO22	I2C clock
Load	Positive	INA219 VIN-	Powered by battery
	Negative	GND	Shared
ESP32	VIN	External 5V	USB/adapter
	GND	GND	Shared

Charge section

Device	Pin	ESP32/Connection	Notes
INA219 (0x41)	VCC	3V3	From ESP32
	GND	GND	Shared
	SDA	GPIO21	I2C data
	SCL	GPIO22	I2C clock

OLED	A0	3V3	Soldered for 0x41
	A1	GND	For 0x41
	VIN+	TP4056 OUT+	Charging path
	VIN-	Battery +/TP4056 B+	To battery
	VCC	3V3	From ESP32
	GND	GND	Shared
	SDA	GPIO21	I2C data
TP4056	SCL	GPIO22	I2C clock
	IN+	5V source	USB/adapter
	IN-	GND	Shared
	OUT+	INA219 VIN+	Charging path
	OUT-	GND	Shared
	B+	INA219 VIN-	To battery +
	B-	GND	Shared
ESP32	VIN	External 5V	USB/adapter
	GND	GND	Shared
Combined			

Device	Pin	ESP32/Connection	Notes
Load	Positive	INA219 (0x40) VIN-	Powered by battery
	Negative	GND	Shared
INA219 (0x40)	VCC	3V3	From ESP32
	GND	GND	Shared
	SDA	GPIO21	I2C data
	SCL	GPIO22	I2C clock

INA219 (0x41)	A0	GND	For 0x40
	A1	GND	For 0x40
	VIN+	Relay NC/Battery +	Discharging path
	VIN-	Load +	To load positive
OLED	VCC	3V3	From ESP32
	GND	GND	Shared
	SDA	GPIO21	I2C data
	SCL	GPIO22	I2C clock
TP4056	A0	3V3	Soldered for 0x41
	A1	GND	For 0x41
	VIN+	Relay NO/TP4056 OUT+	Charging path
	VIN-	Battery +/TP4056 B+	To battery
Relay (5V)	VCC	3V3	From ESP32
	GND	GND	Shared
	SDA	GPIO21	I2C data
	SCL	GPIO22	I2C clock
Relay (5V)	IN+	5V source	USB/adapter
	IN-	GND	Shared
	OUT+	Relay NO	To INA219 (0x41)
	OUT-	GND	Shared
Relay (5V)	B+	INA219 VIN-	To battery +
	B-	GND	Shared
	VCC	VIN/External 5V	5V power
	GND	GND	Shared

	IN	GPIO25	HIGH = charging (NO)
	NO	TP4056 OUT+	Charging path
	NC	Battery +	Discharging path
	COM	INA219 (0x41) VIN+/INA219 (0x40) VIN+	Common
ESP32	VIN	External 5V	USB/adapter
	GND	GND	Shared